

▲ M001 Week 5 — English Worksheet

Topic: First Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

The agent **walks the edges** of a square, dropping blocks as it moves. Stack the squares, shrink each one, and a pyramid grows.

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
  turn left
```

What shape does the agent draw? How many blocks long is each side?

Why is it **repeat 4 times** and **turn left** ? What would **repeat 3 times** draw instead?


```
set f to 6
repeat 3 times:
  [draw one f × f square]
  move up by 1
  change f by -2
```

List the side length of each layer from bottom to top. Looking from the side, what shape do the stacked squares make?


```
place on move ON  
move forward by 4  
place on move OFF  
move forward by 4
```

Where do the blocks end up? Why does turning **place on move** OFF matter when the agent moves up to start the next layer?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to draw a closed **square** outline.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Each new layer must be **2 blocks smaller** than the one below, so the tower narrows into a pyramid.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  move up by 1
  change f by 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each square must sit one block **higher** than the last.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  change f by -2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Now switch to your homework world. Run your `pyra` command to build a pyramid. Then change the **starting value of `f`** (try a smaller number) and build a shorter one. If the agent ends up facing the wrong way or stuck, use your helper commands to reset it — `1` turns it left, `r` turns it right, and `r1` teleports it back to you.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.